

AARAV SHARMA

Machine Learning Engineer | GenAI | MLOps | Backend Systems

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SUMMARY

Machine Learning Engineer with 3+ years of experience building, deploying, and monitoring **ML, deep learning, LLM, and RAG systems** in production. Strong in **Python, PyTorch, scikit-learn, Transformers, Hugging Face, LangChain, vector databases, SQL, FastAPI, Docker, Kubernetes, AWS, MLflow, GitHub Actions**, model evaluation, feature engineering, CI/CD, and inference optimization. Proven record of improving model quality, reducing inference latency, and shipping reliable AI APIs end-to-end.

CORE SKILLS

Languages	Python, SQL, JavaScript, Bash
ML / DL	PyTorch, scikit-learn, TensorFlow, XGBoost, LightGBM, NumPy, pandas, SciPy
GenAI / NLP	LLMs, Transformers, Hugging Face, RAG, embeddings, prompt engineering, fine-tuning, LoRA/PEFT, tokenization, evaluation, tool calling
ML Systems / MLOps	MLflow, model serving, experiment tracking, feature engineering, data pipelines, model monitoring, drift detection, A/B testing, CI/CD
Backend / Data	FastAPI, REST APIs, PostgreSQL, Redis, vector databases, FAISS, pgvector, ETL, Airflow
Cloud / DevOps	AWS (S3, EC2, ECS, ECR, Lambda), Docker, Kubernetes, GitHub Actions, Linux, Git

EXPERIENCE

Machine Learning Engineer - NovaScale AI

Bengaluru, India | Jan 2024 - Present

- Built and deployed **PyTorch** and **scikit-learn** models for ranking, classification, and anomaly detection; improved offline F1 by **14%** through feature engineering, class weighting, hyperparameter tuning, and error analysis.
- Designed a production **RAG** pipeline using **Hugging Face Transformers, embeddings, pgvector, and FastAPI**; improved retrieval Recall@5 by **21%** and reduced hallucination rate by **28%** through chunking, metadata filters, hybrid retrieval, reranking, and evaluation.
- Optimized LLM inference with quantization, batching, caching, and prompt/context reduction; reduced p95 latency by **43%** and cloud inference cost by **31%** while maintaining target quality benchmarks.
- Productionized models with **Docker, Kubernetes, AWS ECS/ECR, MLflow, GitHub Actions**, structured logging, health checks, and automated CI/CD; cut release cycle from 2 days to **3 hours**.
- Implemented model monitoring for data drift, performance degradation, latency, and failure rates using scheduled evaluation jobs and dashboards; reduced undetected model regressions by **40%**.

Machine Learning Engineer - DataForge Labs

Hyderabad, India | Jul 2022 - Dec 2023

- Trained and served **XGBoost, LightGBM, and PyTorch** models for customer propensity and fraud classification; improved ROC-AUC from **0.81 to 0.90** after feature selection, leakage checks, and threshold tuning.
- Built reproducible **Python + SQL** data pipelines processing 5M+ records/day with validation, deduplication, feature generation, and partitioned storage; reduced training-data preparation time by **55%**.
- Developed asynchronous **FastAPI** inference services with Redis caching, PostgreSQL persistence, authentication, rate limiting, request validation, and observability; sustained **300+ requests/sec** under load tests.

SELECTED PROJECTS

Enterprise RAG Copilot | Python, PyTorch, Transformers, pgvector, FastAPI, Docker

- Built a document-grounded question-answering system over 100K+ chunks; implemented hybrid retrieval, reranking, citation generation, evaluation sets, caching, and confidence thresholds.

Real-Time Vision Anomaly Detector | PyTorch, OpenCV, ONNX, AWS

- Trained a CNN-based visual anomaly detector and exported optimized inference with ONNX Runtime; reduced CPU inference latency by 47% and added drift/evaluation monitoring.

ML Platform / Experiment Tracker | Python, MLflow, PostgreSQL, Airflow, Docker

- Created reusable training pipelines with versioned datasets, experiment tracking, model registry, scheduled retraining, validation gates, and rollback-ready deployments.

EDUCATION

B.Tech / B.E. in Computer Science and Engineering - National Institute of Technology, India | 2022

CERTIFICATIONS & TECHNICAL KNOWLEDGE

Deep Learning Specialization | AWS Certified Machine Learning - Specialty | Strong understanding of supervised/unsupervised learning, probability, statistics, linear algebra, optimization, distributed inference, data structures, algorithms, system design, API design, and software engineering best practices.

ATS KEYWORDS

Machine Learning Engineer, ML Engineer, AI Engineer, Applied Scientist, NLP, Computer Vision, Generative AI, LLM, RAG, Retrieval-Augmented Generation, Transformer, attention, embeddings, vector database, fine-tuning, LoRA, PEFT, PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, Python, SQL, FastAPI, REST API, PostgreSQL, Redis, FAISS, pgvector, Docker, Kubernetes, AWS, MLflow, Airflow, GitHub Actions, CI/CD, model serving, model monitoring, MLOps, feature engineering, A/B testing, inference optimization, data pipelines, cloud deployment.

ATS stress-test sample resume. Company names, metrics, and contact details are fictional.